

Priyanka Magar

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RESEARCH INTERESTS

My research interests include geometric topology, stable homotopy theory, and K-theory. My doctoral research focused on the classification of smooth structures on higher-dimensional manifolds, particularly their classification up to isotopy in dimensions 8 to 10.

ACADEMIC EXPERIENCE

Research Associate Indian Institute of Technology Madras	12 July, 2024 – Present
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Research Associate Indian Institute of Technology Bombay	28 May, 2024 – 11 July, 2024
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EDUCATION

Indian Institute of Technology Bombay , Mumbai, India Doctor of Philosophy Thesis: Smooth structures on PL-manifolds of dimensions between 8 and 10.	July 2018 — May 2024 Defense date - 22 Nov 2024 Cumulative GPA: 7.96
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University of Pune , Pune, India Masters of Science in Mathematics	July 2015 — May 2017 Cumulative GPA: 8.05
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S. P. College (Aff. University of Pune) , Pune, India Bachelor of Science in Mathematics	July 2013 — May 2015 Percentage: 87.25%
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PUBLICATIONS/PREPRINTS

- **P. Magar-Sawant**, *Concordance structure set of connected sum of projective spaces*.
<https://doi.org/10.1007/s12044-024-00800-7>. Proceedings-Mathematical Sciences
- S. Basu, R. Kasilingam, **P. Magar-Sawant**, *Smooth structures on PL-manifolds of dimensions between 8 and 10*. <https://arxiv.org/abs/2302.02301>. Positive review with Minor revisions.
- **P. Magar-Sawant**, *Tangential smoothings of k -fold connected sum of complex projective spaces*,
<https://arxiv.org/abs/2306.16911>. Positive review with Minor revisions.

CONFERENCES/WORKSHOPS/TALKS

Oral Presentation

- “Concordance structures of sets of 8 to 10-dimensional manifolds”, *Young Topologists Meet-2024 Münster*, Germany, August 2024.
- “Smooth structures on connected sum of projective spaces”, *International Conference on Topology and its Application-2023 Nafpaktos*, Greece, July 2023.
- “On the stable cohomotopy and KO groups of connected sum of projective spaces”, *37th Annual Conference of the Ramanujan Mathematical Society*, Chennai, December 2022.

Workshop

- Operads in Topology, NCM Workshop, IIT Bombay, India, Nov 2024.
- Hochschild Homology, NCM Workshop, IISER Kolkata, India, Dec 2023.
- Cohen Macaulay simplicial complexes in graph theory, NCM Workshop, Chennai Mathematical Institute, India, July 2023.
- Dualities in Topology and Algebra, International Centre for Theoretical Sciences, TIFR Bengaluru, May 2023.
- Dualities in Topology and Algebra(Online), International Centre for Theoretical Sciences, TIFR Bengaluru, Feb 2021.
- Global Homotopy Theory, Department of Mathematics, IIT Bombay, Feb 2020.
- Diamond Jubilee Symposium, Department of Mathematics, IIT Bombay, Jan 2019.
- International conference on Algebra and Analysis, Department of Mathematics, SPPU, Pune, Dec 2017.
- Annual Foundation School-II, IISER Pune, 2018.
- Annual Foundation School-I, Bhaskaracharya Pratishthana, Pune, 2017.

Technical Talks

- “Classifying Spaces of Categories”, *Topology and Related Topics Seminar*, IIT Bombay, Feb 2024.
- “Quillen equivalent stable homotopy categories”, *Topology and Related Topics Seminar*, IIT Bombay, August 2023.
- “A brief introduction to the stable homotopy category”, *Topology and Related Topics Seminar*, IIT Bombay, August 2023.
- “Essential Categorical Terms for Stable Homotopy Theory”, *Topology and Related Topics Seminar*, IIT Bombay, August 2023.
- “Introduction to Spectra and Stable Homotopy Category”, *Dualities in Topology and Algebra 2023*, ICTS Bangalore, May 2023.
- “Spectra and stable homotopy category” in tt-category and chromatic homotopy theory online seminar series, March 2023.

Expository Talks

- “Simplicial sets arising from categories” at Student Seminar 37, September 2023, IIT Bombay.
- “A brief introduction to the fiber bundle” at Student Seminar 19, April 2023, IIT Bombay.

Teaching Assistantship

Institute

- Numerical Analysis – MA 214, Spring 2021–22
- Differential Equations – MA 207, Autumn 2021–22
- Linear Algebra – MA 106, Spring 2020–21

- Calculus II – MA 111, Spring 2020–21
- Differential Equations – MA 207, Autumn 2020–21
- Calculus I – MA 105, Autumn 2019–20

NPTEL

- Introduction to Algebraic Topology-Part II: Prof. Anant Shastri, Autumn 2021
- Linear Algebra: Dr. Dilip Patil. Autumn 2024.

AWARDS/SCHOLARSHIPS/FELLOWSHIPS

UGC NET. AIR 12

UGC JRF. AIR 78

SKILLS

- **Environment:** $\text{\LaTeX}$, Microsoft Office
- **Operating System :** Windows, Linux
- **Computer Algebra Systems :** Python, SageMath